

Dheeren Tejani

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Portfolio: dheerentejani.netlify.app | **Location:** Mumbai, India

Summary

AI Engineer focused on Generative AI and efficient ML systems. Experienced in building and optimizing LLMs and diffusion models from scratch, and deploying production-grade inference pipelines, reducing costs near **7x** through model quantization and hardware-aware optimization.

Experience

AI Engineering Intern

Ally Blocksecure Media Pvt. Ltd. (TheAlly)

Jan 2026 – Present

- Built an end-to-end PDF parsing pipeline, replacing a 70B LLaMA-based system with a 4-bit quantized 7B Qwen model, achieving **10x parameter reduction** and cutting inference costs by nearly **7x** (2.67 USD/hr on H100 → 0.40 USD/hr on L4). Developed a FastAPI backend with async job queuing and encryption, handling the full pipeline from document preprocessing to structured output postprocessing.

Full Stack Developer

Vivekanand Education Society (VESASC)

Jul 2025 – March 2026

- Developed the official **Exam Remuneration System for both written and practical examinations** (Serverless), digitalizing workflows for **100+ faculty**, replacing manual paper processes with a centralized, secure online platform.

Projects

LLaMa style Large Language model

PyTorch, Transformers, React

- Built a 160M-parameter decoder-only Transformer from scratch in PyTorch (self-attention, positional embeddings, causal masking). Pretrained on Cosmopedia dataset over **84K steps / 22B tokens at 400K tokens/sec** on H100 using BF16 AMP, gradient accumulation, cosine LR scheduling, and FlashAttention. Model demonstrated **instruction-following capability** after OpenHermes SFT, achieving a final loss of **1.5**. Hosted it with Docker and FastAPI Backend and React Frontend

Diffusion Transformer (DiT) — Image Generation model

PyTorch, Flow Matching, RoPE

- Built a **1.6B-parameter** Diffusion Transformer from scratch with flow matching, 2D RoPE positional encodings, Heun sampler, and timestep shift. Pre-latentized **6M Danbooru images** using Flux's 16-channel VAE across 5 aspect ratio buckets. Trained for **14K steps** on B200 GPU, achieving a final MSE loss of **0.50** (lowest 0.48). Model successfully learned global color distribution, contrast, and style coherence.

Skills

Languages: Python, JavaScript/TypeScript, SQL

AI/ML: PyTorch, Computer Vision, NLP, Model Optimization, Flow Matching, MLOps, Quantization

Deployment: RunPod, FastAPI, Streamlit, Docker, Git, AWS

Web Dev: React, TypeScript, Tailwind CSS

Data/Tools: NumPy, Pandas, OpenCV, Matplotlib

Education

B.Sc. in Artificial Intelligence

Jun 2023 – Apr 2026

College : Vivekanand Education Society's College of Arts, Science and Commerce (Autonomous) Mumbai

CGPA : 8.65